

Modulation of Hepatic Drug Metabolizing Enzymes by Dietary Doses of Thymoquinone in Female New Zealand White Rabbits

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Herbal medicines can affect drug metabolizing enzymes. Therefore the effect of thymoquinone (TQ), the active ingredient of black seeds, was examined on rabbit liver drug metabolizing enzymes. Two groups of New Zealand female rabbits received TQ at 10 and 20 mg/kg/day orally and a control group of six animals each were killed after 8 weeks. Blood and livers were harvested and the activity of cytochrome P450 (CYP) and phase II enzymes in the microsomal and cytosolic preparations were measured by HPLC and ELISA methods. The liver enzymes ALT/AST and albumin were similar in the three groups. CYP1A2, CYP3A4, but not CYP2E1, were significantly diminished by TQ treatment. Of the phase II enzymes, glutathione-S-transferase (GST) and glutathione peroxidase (GPx) were significantly induced by the high TQ dose, while the total glutathione levels were unaffected. Glutathione reductase (GR), on the other hand, was significantly induced in the two experimental groups. Thymoquinone has differential effects on CYP and phase II enzymes. Inhibition of some CYP enzyme activities may lead to a hazardous herb–drug interaction. Induction of GR activity may explain the salutatory effect of the black seeds in inhibiting the generation of bioactive metabolites known to promote carcinogenesis and oxidative cell damage. Copyright © 2012 John Wiley & Sons, Ltd.

Keywords: herbal remedies; drug metabolism; CYP enzymes; thymoquinone; black seed; *Nigella sativa*.

INTRODUCTION

Conventionally, drug metabolism takes place by two consecutive processes, phase I and phase II (Glue and Clement, 1999). Phase I reactions are principally mediated by CYP enzymes that are mainly expressed in the liver and catalyse oxidation, reduction and hydrolysis of the drug to generate a more polar and rapidly eliminated metabolites (Glue and Clement, 1999). The principal human CYP isoforms that are involved in drug metabolism include CYP1A1/2, CYP2B6, CYP2C8/9/19, CYP2D6, CYP2E1 and CYP3A4/5. Although CYP enzymes normally generate metabolites with diminished biological activity, there are numerous examples in which these enzymes mediate the formation of reactive intermediates from chemically inert agents (Shimada *et al.*, 1994). These reactive metabolites play a key step in initiating cellular damage and cancers.

Phase II reactions mediate the conjugation of exogenous and endogenous compounds as well as reactive metabolites produced during phase I metabolism with molecules such as uridine 5'diphosphate-glucuronic acid, glutathione or sulfate to produce readily eliminated metabolites (Brand *et al.*, 2010).

For more than 2000 years, herbal remedies have remained useful for treating a large number of diseases.

Several and severe herb–drug interactions are well known to occur in patients who combine particular herbal and dietary supplements with their medications (Elbarbry *et al.*, 2008). Components of herbal extracts can affect metabolism/elimination of conventional pharmaceuticals through the modulation of the activity of cytochrome P450 (CYP) enzymes in the absence of changes in liver enzymes such as ALT/AST. Such modulation of CYP enzymes could potentially affect therapeutic outcomes and can have significant long-term effects (Guengerich, 1997). Reports of a further increased public interest in herbal medicines warrants, therefore, a careful examination of the potential hazards associated with their use.

The seeds of the black seeds, or *Nigella sativa*, have long been used in folk medicine for a wide range of illnesses including allergy, bronchial asthma, headache, dysentery, infections, obesity, back pain, hypertension and gastrointestinal problems (Al-Rowais, 2002; Al-Ghamdi, 2001). Additionally, the active component of black seeds, thymoquinone (TQ), reduces oxidative stress and the probability of developing cancer (Fujita *et al.*, 2005; El *et al.*, 2006). Its safety measured by normal liver enzymes was reported (El-Dakhakhny, 1965), while its effect on CYP enzymes has not been documented to our knowledge.

Due to the widespread consumption of *Nigella sativa* seed and its oil extract, it was of interest to determine the effect of its principal active compound, TQ, on drug metabolizing enzymes above and beyond the routine liver enzymes, ALT/AST. It was reasoned that the reported normal liver enzymes in previous studies do not preclude

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the presence of a potentially harmful effect of TQ on the drug metabolizing enzymes. With this goal in mind, this study aims to investigate the potential modulatory effects of thymoquinone (TQ) on hepatic drug metabolizing enzymes in female New Zealand White rabbits.

MATERIALS AND METHODS

Medications. Thymoquinone: 2-Isopropyl-5-methyl-1,4-benzoquinone (Sigma-Aldrich, St Louis, MO, USA) was dissolved in pure corn oil (Sigma-Aldrich, St Louis, MO, USA) to give a concentration of 75 mg/mL.

Animals and experimental design. This study was conducted on 18 female New Zealand White rabbits. The animals were obtained from Charles River Laboratories, Inc. (Wilmington, MA, USA). All the rabbits were 6–8 weeks old weighing between 1.5 and 2.0 kg at the beginning of the study. The animals were housed as per the University of Saskatchewan regulations in the Health Sciences Building and all procedures followed the animal assurance quality guidelines. The review Board of the University granted ethics approval for this study. The rabbits were housed in cages at room temperature of $22 \pm 2^\circ\text{C}$ and a relative humidity of 40% to 60% under a 12 h light/dark cycle to approved standards for laboratory animal care. At the end of the 8 weeks, the animals were killed according to the current animal care regulations. The rabbits were randomly divided into three groups of six animals each. The first group served as the control, whereas groups 2 and 3 were provided with TQ (10 mg/kg and 20 mg/kg, orally once daily) respectively for 8 weeks. Regular diet and water were provided *ad libitum*.

On arrival, all the rabbits were left for 7–14 days for acclimatization. Thereafter, the rabbits received the designated TQ doses as mentioned above. At the end of the 8 weeks and after 20 h fasting, they were anaesthetized by injecting ethanol, pentobarbital sodium, 50 mg/kg (Bimeda-MTC Animal Health Inc. Cambridge, ON, Canada) in the marginal ear vein. Blood samples were collected and were used for measurements of the fasting serum transaminases and albumin as described previously (Ragheb *et al.*, 2009). Livers were rapidly removed and rinsed in ice-cold 0.9% NaCl, and were flash frozen in liquid nitrogen and stored at -80°C until use.

Chemicals and reagents. *p*-Nitrophenol (PNP), *p*-nitro-catechol (4NC), midazolam (MDZ), 4-hydroxymidazolam (4 OH-MDZ), resorufin, methoxyresorufin and all chemicals used for microsomal and cytosolic preparation, determination of protein content and enzymatic assays were purchased from Sigma-Aldrich Canada Ltd (Oakville, ON, Canada). Acetonitrile and methanol were high pressure liquid chromatography (HPLC) grade obtained from EMD Chemicals (Durham, NC). All other chemicals used were analytical grade.

Preparation of hepatic microsomes and cytosols. Hepatic microsomes and cytosols were prepared as described previously (Elbarbry *et al.*, 2007). Briefly, 0.5 g liver samples were accurately weighed and homogenized in 2 mL ice-cold homogenization buffer (50 mM Tris buffer, 150 mM KCl, 0.1 mM dithiothreitol, 1 mM ethylenediamine tetra-acetate (EDTA), 20% glycerol and

0.1 mM phenylmethylsulfonylfluoride) using a Polytron homogenizer. The homogenate was centrifuged at $9000 \times g$ for 30 min at 4°C in a Beckman L8-55 Ultracentrifuge (Palo Alto, CA). The supernatant was carefully transferred to clean ultracentrifuge tubes and centrifuged at $100000 \times g$ for 30 min at 4°C . The supernatant (the cytosol) was removed and transferred to labelled tubes and stored at -80°C until use. The pellet was washed in 2 mL of ice-cold 150 mM KCl and centrifuged again at $100000 \times g$ for 30 min at 4°C . The pellet was resuspended in 2 mL of ice-cold 0.25 M sucrose solution and aliquots were stored in cryogenic microcentrifuge tubes (400 μL) at -80°C until use.

Determination of microsomal and cytosolic protein content. Protein concentrations of microsomal and cytosolic fractions were determined in triplicate by the method of Lowry *et al.* (1951) using bovine serum albumin as a calibration standard. Absorbance was measured at 750 nm on an Agilent 8453E UV-visible spectrophotometer using Chemstation software (Palo Alto, CA).

Measurement of CYP (1A2, 2E1 and 3A4) activity in hepatic microsomes. Preliminary experiments in rabbit liver microsomes were conducted to determine the optimum incubation conditions (incubation time and microsomal protein content) to give linear metabolite formation kinetics for CYP2E1-mediated hydroxylation of PNP, CYP1A2-mediated methoxyresorufin-O-dealkylation (MROD) and CYP3A4-mediated hydroxylation of MDZ. Metabolite formation rate was calculated by dividing the amount of the metabolite formed by the incubation time and microsomal protein content (nmol/min/mg).

PNP hydroxylase assay. Metabolite formation velocities for PNP were measured at a saturating probe substrate concentration of $150 \mu\text{M}$ in rabbit liver microsomes to provide estimates of V_{max} . The PNP hydroxylase activity was determined by quantification of 4NC formation rates in rabbit liver microsomes using a standard protocol as described before (Elbarbry *et al.*, 2007). Briefly, the incubation mixtures consisted of PNP ($150 \mu\text{M}$), 0.4 mg/mL liver microsomal protein, 1 mM ascorbic acid, 2 mM MgCl_2 , 1 mM NADPH and 50 mM phosphate buffer, pH 6.8, in a final volume of 0.5 mL. After a preincubation period of 1 min at 37°C , the reaction was started by addition of NADPH and incubated at 37°C for 30 min in a shaking water bath. The reaction was terminated by the addition of 50 μL ice-cold phosphoric acid and processed as described elsewhere (Elbarbry *et al.*, 2006).

MROD assay. Incubation mixtures contained 50 mM potassium phosphate buffer (pH 7.4), 2 mM MgCl_2 , 0.08 mg/mL liver microsomal protein, methoxyresorufin (500 nM) and 1 mM NADPH, in a final volume of 0.5 mL. The samples were preincubated for 1 min and the reaction was started by the addition of NADPH. The mixture was incubated at 25°C for 8 min with shaking. The reaction was terminated by placing the tubes on ice and adding 0.5 mL ice-cold methanol with vortex mixing. After cooling on ice for 15 min, the mixture was centrifuged at $6000 \times g$ for 20 min. The supernatant was filtered with a membrane filter of 0.45 μm pore size. Fifty microliters of the filtrate was injected directly into the analytical column for immediate HPLC analysis as described previously (Elbarbry *et al.*, 2007).

MDZ hydroxylase assay. The MDZ hydroxylase activity was determined by quantification of the 4-OH MDZ formation rate in rabbit hepatic microsomes. Microsomal incubation mixtures consisted of MDZ (100 μ M), 0.4 mg/mL liver microsomal protein, 2 mM $MgCl_2$, 1 mM NADPH and 50 mM phosphate buffer, pH 7.8, in a final volume of 0.5 mL. After a preincubation period of 1 min at 37 °C, the reaction was started by addition of NADPH and incubated at 37 °C for 30 min in a shaking water bath. The reaction was terminated by addition of 50 μ L ice-cold acetonitrile and processed as described before (Elbarbry *et al.*, 2009).

Measurement of the phase II enzyme activity in hepatic cytosols. Total glutathione content was evaluated by the enzymatic recycling procedure using glutathione reductase and 5,5'-dithio-bis (2-nitrobenzoic acid) (Griffith, 1980). General glutathione-S-transferase (GST) activity was monitored by spectrophotometric analysis of glutathione conjugation of 1-chloro-2,4-dinitrobenzene (CDNB). Glutathione reductase (GR) and glutathione peroxidase (GPx) activities were determined in liver cytosols using the GR assay kit of Oxford Biomedical Research (Oxford, MI, USA) and the total GPx assay kit from ZeptoMetrix Corporation (Buffalo, NY, USA), respectively, following the manufacturer's protocol.

Statistical analysis. Descriptive statistics (mean $\pm$ SD) were used to report the activity within the same group. Significant differences in activity pattern between treatments groups relative to the control group were assessed with one-way analysis of variance, with Tukey's test for pairwise multiple comparisons using Prism 5 (GraphPad software, San Diego, CA). The level of significance was set at $p < 0.05$.

RESULTS

Effects of selected TQ doses on liver function

Exposure to the selected doses of TQ had no obvious effect on the animal behaviour. There was a progressive increase in the body weight of all the groups. No significant differences were found between the groups at the beginning, throughout, and at the end of the 8-week period (data not shown). Table 1 shows the effect of TQ treatment on liver enzymes, expressed as aspartate transaminase (AST), alanine transaminase (ALT) and the albumin levels. No significant differences were found in AST, ALT or albumin levels.

Table 1. Effect of treatment with thymoquinone on serum transaminases and albumin levels

Liver enzyme/protein	Control	TQ (10 mg/kg)	TQ (20 mg/kg)
AST (U/L)	36.0 $\pm$ 6.8	33.3 $\pm$ 15.1	59.0 $\pm$ 27.3
ALT (U/L)	41.2 $\pm$ 10.4	40.0 $\pm$ 25.0	48.7 $\pm$ 35.0
Albumin (g/L)	17.5 $\pm$ 1.9	16.0 $\pm$ 1.0	15.8 $\pm$ 1.8

Effect of TQ treatment on rabbit CYP450 activities

To assess the effect of TQ on CYP1A2, CYP2E1 and CYP3A4 maximal enzyme activities (V_{max}), hepatic microsomes from rabbits treated with different doses of TQ were incubated with saturating probe substrate concentrations. Hepatic CYP2E1 activity, as indicated by PNP hydroxylation, was not significantly affected by the treatment with TQ, although a trend of dose-dependent inhibition was observed (Fig. 1). More modest decreases were observed in the O-dealkylation of methoxyresorufin (MROD), a marker of CYP1A2 activity, and hydroxylation of midazolam, a marker of CYP3A4 activity. However, statistical significance was only attained at the higher TQ dose (20 mg/kg) as illustrated in Fig. 1.

Effect of TQ treatment on rabbit phase II enzymes activities

To determine the effects of TQ treatment on liver phase II enzymes, activities of glutathione peroxidase (GPx), glutathione reductase (GR), glutathione-S-transferase (GST) and total content of glutathione (GSH) were assessed in liver cytosols. Treatment of rabbits with TQ resulted in a trend of dose-dependent induction in the GST activity in the liver determined using 1,2 dichloro-2,4-dinitrobenzene (CDNB) as a substrate. However, a significant level of induction was only observed in the group treated with the high dose of TQ (20 mg/kg) (Fig. 2). A similar pattern of induction, a dose-dependent increase with only significant changes observed after treatment with 20 mg/kg dose, was also observed in GPx activity. Although total glutathione levels were unaffected by the TQ treatment, GR activity was induced at both dose levels (10 and 20 mg/kg) (Fig. 2). No significant difference in GR activity was observed between the 10 and 20 mg/kg treatments.

DISCUSSION

Thymoquinone, TQ, is the major constituent of the volatile oil of black seed (*Nigella sativa*), which has a variety of beneficial properties including hepatoprotective, chemopreventive, antibacterial and antiinflammatory activities. Its mechanism of action has been ascribed, at least partly, to its ability to scavenge free radicals and/or inhibit lipid peroxidation (Ragheb *et al.*, 2009). Although the public use of black seed oil extract is increasing, it is surprising that the effect of oral intake of dietary doses of TQ, the

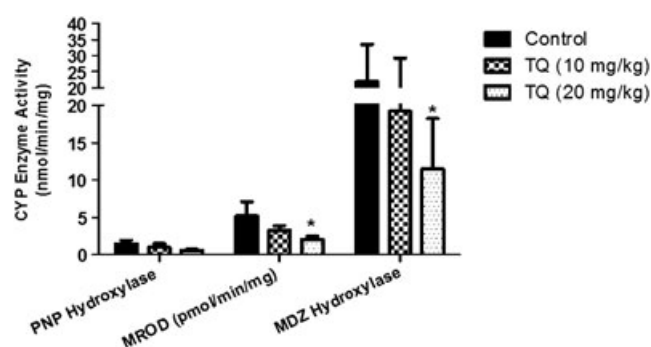


Figure 1. Effect of thymoquinone (TQ) treatment on the activities of hepatic CYP2E1, CYP1A2 and CYP3A4 in rabbits.

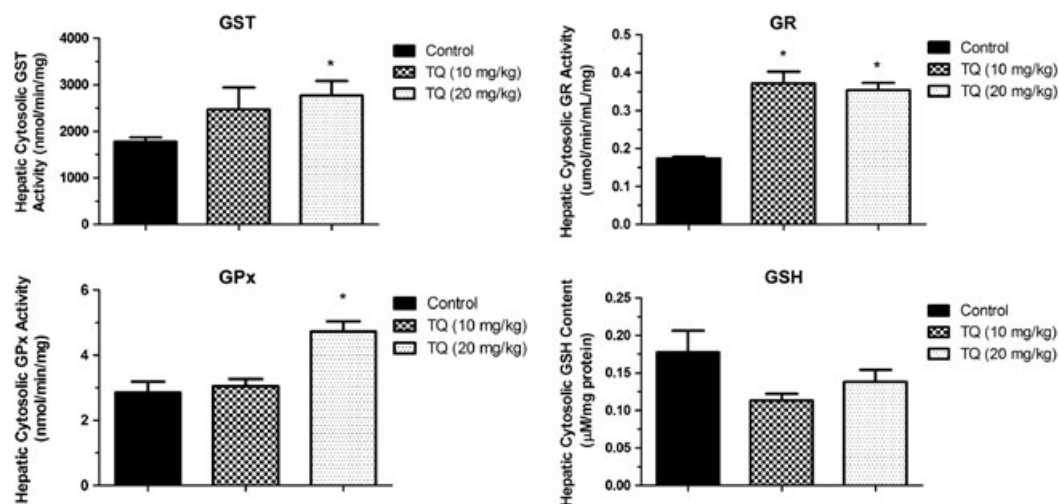


Figure 2. Effect of thymoquinone (TQ) treatment on the activities of hepatic phase II enzymes in rabbits.

major constituent of the oil, on drug metabolizing enzymes has not been investigated. This was the principal objective of this study.

The lower TQ dose employed in this study was chosen to reflect a human intake, equivalent to 40 mg/kg/day *Nigella sativa* oil (Kalus *et al.*, 2003; Goreja, 2003). An additional group of rabbits was treated with a higher dose of TQ to evaluate whether any effects on the drug-metabolizing enzymes were dose-dependent.

The study examined the effect of TQ on selected CYP enzymes with a well-documented potential role in drug interactions or drug/toxin-induced toxicity/cancer. For example, CYP3A4 is involved in the metabolism of more than 50% of drugs available on the market (Glue and Clement, 1999). CYP2E1 catalyses the bioactivation of important environmental chemicals, bioactivates a selected number of drugs (i.e. acetaminophen, industrial solvents (i.e. carbon tetrachloride, CCl_4) (Brade *et al.*, 1986) and environmental pro-carcinogens (i.e. nitrosamines) to highly reactive intermediates (Elbarbry *et al.*, 2007). These reactive metabolites can form adducts with endogenous biomolecules such as DNA, RNA and proteins, and play a key role in initiating cellular damage and cancers. The crucial step in the carcinogenesis of polycyclic aromatic hydrocarbons (PAH) is their biotransformation to the oxyderivatives by CYP1A isoforms (Wassenberg *et al.*, 2005; Watson *et al.*, 1998; Cheung *et al.*, 1994).

It was observed that TQ modulated the activity of CYPs in an isoform-specific manner. The CYP3A4-mediated hydroxylation of midazolam was inhibited, but was also significant at the higher of the two doses used. Inhibition of CYP activity by TQ could precipitate severe drug interactions especially with drugs metabolized by CYP3A4 to a large extent (Table 2). Similarly, the O-dealkylation of methoxyresorufin, a marker for CYP1A2 isoform, was also inhibited by treatment with the higher TQ dose. CYP1A2 is an important enzyme in the bioactivation of chemical carcinogens, such as aromatic and heterocyclic amines. It is possible, then, that the suggested cancer chemopreventive effect of TQ is caused by inhibition of the generation of bioactive metabolites. On the other hand, treatment with TQ failed to modulate CYP2E1-mediated hydroxylation of PNP. Several studies have documented the protective effect of TQ against CCl_4 -induced hepatotoxicity (Al-Ghamdi, 2003).

Because CYP2E1 is responsible for the bioactivation of CCl_4 to a hepatotoxic metabolite (Qin *et al.*, 2005), our results suggest that mechanisms other than suppression of CYP2E1, the bioactivation pathway, are modulated by TQ and contribute to the reported TQ-protective role against CCl_4 -induced hepatotoxicity.

The study then evaluated the effect of TQ on phase II detoxification system. Glutathione (GSH) and its related enzymes, namely glutathione-S-transferases (GST), glutathione reductase (GR) and glutathione peroxidase (GPx) are key players in the detoxification of reactive metabolites and the prevention of cellular oxidative damage (Elbarbry and Alcorn, 2009). To evaluate this hypothesis, the activity of phase II enzymes were examined in hepatic cytosols from TQ-treated rabbits. Treatment with TQ failed to influence total (reduced and oxidized) GSH level but, both TQ doses elevated GR; an enzyme that maintains glutathione in its reduced form that detoxifies reactive metabolites. Additionally, cytosolic GST (a family of detoxification enzymes that conjugate reduced GSH to a wide variety of xenobiotics and reactive intermediates (Hayes *et al.*, 2005), and protect vital cellular nucleophilic constituents against electrophilic reactive species and prevent cellular damage) was elevated

Table 2. Clinically significant drugs that are metabolized by CYP3A4 and the potential consequences of CYP3A4 inhibition after co-administration with thymoquinone

Drug substrate	Effect of CYP3A4 inhibition
Alprazolam	Excessive sedation and impaired psychomotor function
Atorvastatin, Lovastatin	Increasing the risk of severe myopathy or rhabdomyolysis
Cyclosporine	Increasing the risk of nephrotoxicity
Verapamil	Increasing risk of cardiotoxic reactions include hypotension, bradycardia, cardiac conduction defects (e.g. junctional rhythm with atrioventricular dissociation and high-degree atrioventricular block, including asystole) and arrhythmias
Warfarin	Increasing risk of bleeding

in the higher TQ dose group. A similar pattern was also observed with GPx, an enzyme that catalyses the reduction of hydroperoxides using reduced GSH and functions to protect cells from oxidative damage (Mathers *et al.*, 2004). Taken together, TQ enhanced the ability to eliminate the toxic metabolites.

The importance of these findings are emphasized by the reported association between the presence of certain phytochemicals in grapefruit juice and the decreased rate of metabolism of Ca²⁺-channel blockers used in the treatment of hypertension. Such decreases in metabolism can explain the phenomenon of hypotension associated with increasing drug concentrations (Elbarbry *et al.*, 2008). With the case of grapefruit juice, flavonoids such as naringen down-regulate the expression of cytochrome P450 3A4 (CYP 3A4) in intestinal epithelium (Lown *et al.*, 1998). Another example is St John's wort, widely used for depressive disorders, is a potent inducer for CYP3A4 and accordingly enhances the elimination and decreases the therapeutic effect of several antibiotics and anti-HIV drugs.

CONCLUSION

In conclusion, the results demonstrate that dietary doses of TQ can modulate the hepatic activity of xenobiotic-metabolizing enzyme systems in rabbits. Induction of phase II enzymes and/or inhibition of CYP1A2 explain the chemoprotective effect of TQ against toxins/carcinogens that require bioactivation by CYP1A2. Induction of phase II enzymes can also explain the protective effect of TQ against xenobiotics that require bioactivation through CYP2E1. The significant inhibition of CYP3A4 by TQ should raise a concern regarding the potential herb–drug interaction especially medications that are metabolized by CYP3A4.

Conflict of Interest

The authors declare that there are no financial or commercial conflict of interest.

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